



MATHEMATICS EXTENSION 2

2010 HSC Assessment Task 1
November 2009

Time Allowed - 90 minutes
(plus 5 minutes reading time)

Topics: Circular Motion (Exercises 1-8), Curve Sketching (Exercises 1-8)

General Instructions:

- There are TEN (10) Questions which are NOT of equal value.
- Questions do not necessarily appear in order of difficulty.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number clearly at the top of each question and clearly number each question.
- Use $g = 10 \text{ ms}^{-2}$

Total: 90 marks

QUESTION 1 (8 Marks)

MARKS

Sketch the following on separate number planes, showing important features:

- | | |
|--|---|
| a. $y = (x+1)^2(x-2)(x-4)^3$ | 2 |
| b. $x = \sqrt{4-y^2}$ | 2 |
| c. $\frac{x^2}{16} + \frac{y^2}{25} = 1$ | 2 |
| d. $x^2 - y^2 = 4$ | 2 |

QUESTION 2 (12 Marks)

MARKS

a. Given $F(x) = 1 - x^2$, sketch:

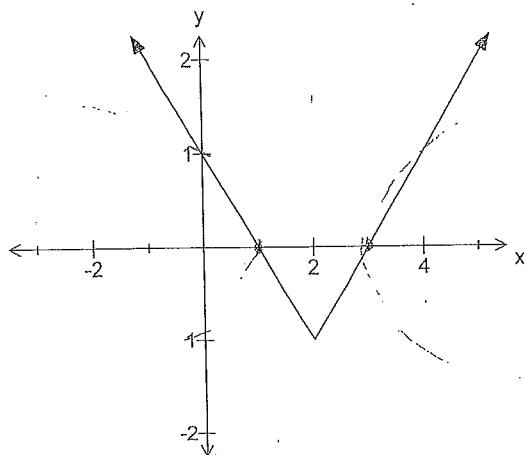
- | | |
|----------------------|---|
| i. $y = F(x)$ | 1 |
| ii. $y = -F(x)$ | 1 |
| iii. $y = F(x) $ | 2 |
| iv. $y = \{F(x)\}^2$ | 2 |

b. For the following, sketch $y = f(x)$, $y = f^{-1}(x)$ and $y = x$ on the same number plane. Show any key points, including all intersection points.

- | | |
|------------------|---|
| i. $f(x) = x^3$ | 3 |
| ii. $f(x) = 2^x$ | 3 |

QUESTION 3 (7 Marks)

MARKS



Copy the function $y = f(x)$ as shown above and sketch on the same number plane :

a. $y = f(|x|)$

2

b. $y = \frac{1}{f(x)}$

2

c. $y^2 = f(x)$

2

d. $y = f(x-2)$

1

QUESTION 4 (10 Marks)

MARKS

Sketch the following on separate number planes, showing important features :

a. $y = \frac{3x+4}{x+2}$

2

b. $25y^2 - 9x^2 = 225$

2

c. $y = x - \frac{2}{x+1}$

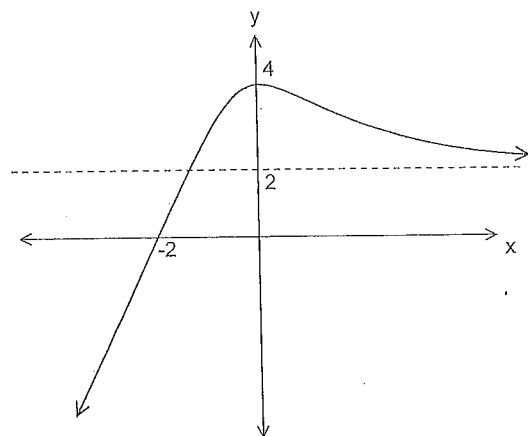
3

d. $y = \frac{3}{(x+2)(x-1)^3}$

3

QUESTION 5 (8 Marks)

MARKS



Copy the function $y = f(x)$ as shown above and sketch on the same number plane :

a. $y = f(-x)$

1

b. $y = 2^{f(x)}$

2

c. $y = -\sqrt{f(x)}$

2

d. $y = f^{-1}(x)$

2

e. $|y| = f(x)$

1

QUESTION 6 (8 Marks)

MARKS

A 3 metre piece of string AB has a mass of 5 kg attached at point B. The string is rotated in a horizontal circle about A and breaks as soon as it exceeds a speed of rotation of 45 revolutions per minute.

a. Find the maximum possible tension in the string.

3

b. The mass at B is replaced by a 3 kg mass and an additional 1 kg mass is attached to the string at C, 2 metres from A (as shown below). Find the new maximum number of revolutions per minute that the string can be rotated.

5

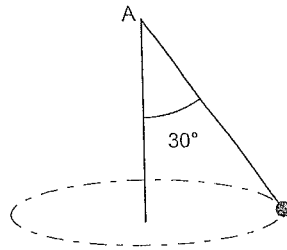


QUESTION 7 (8 Marks)

MARKS

A mass is rotating in circular motion as a conical pendulum as shown in the diagram.

The mass of 5 kg is attached to A by a light string of length 2 m and the angle between the string and the vertex is 30° .



- a. Find the tension in the string.
- b. Find the speed that the mass is moving. Give your answer in m/s.

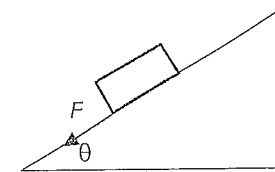
4

4

QUESTION 8 (10 Marks)

MARKS

A section of a car-racing track is being constructed and is to be banked at an angle of θ as shown below. A car of mass m kg is travelling around the track at v m/s.



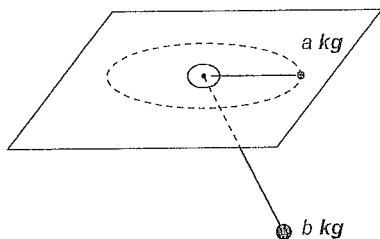
- a. Copy the diagram and mark in the normal force N and the gravitational force. 1
- b. Resolve the forces vertically and horizontally. 2
- c. Hence, find an expression for the : 4
 - i. lateral force F in terms of m, v, r, g and θ .
 - ii. normal force N in terms of m, v, r, g and θ .
- d. The road is constructed around a curve of radius 150 m to allow for an optimum speed of 120 km/h. Calculate the angle (answer to the nearest minute) at which the track should be banked. 3

QUESTION 9 (7 Marks)

MARKS

Two masses are connected by a light inelastic string. One mass of a kg is rotating in circular motion on a smooth horizontal table with angular velocity ω rad/s. The other mass of b kg is at the other end of the string, rotating in a conical pendulum below the table with angular velocity 2ω rad/s.

If $a:b = 5:2$, find what fraction of the string lies above the table.

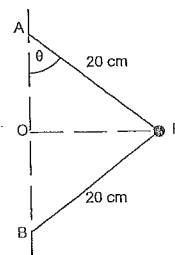


7

QUESTION 10 (12 Marks)

MARKS

A particle, P, of mass m , is attached by two strings, each of length 20 cm, to two fixed points, A and B, which are 24 cm apart and lie on a vertical line as shown in the diagram.



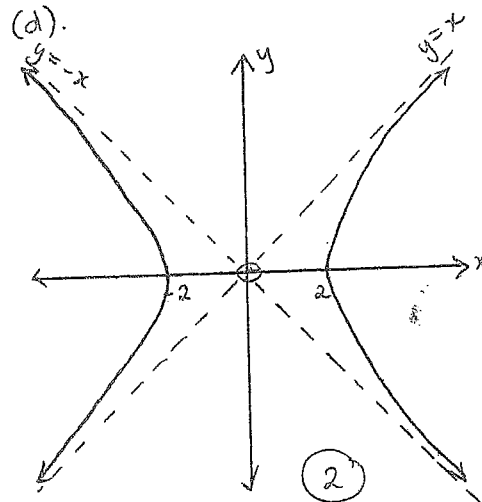
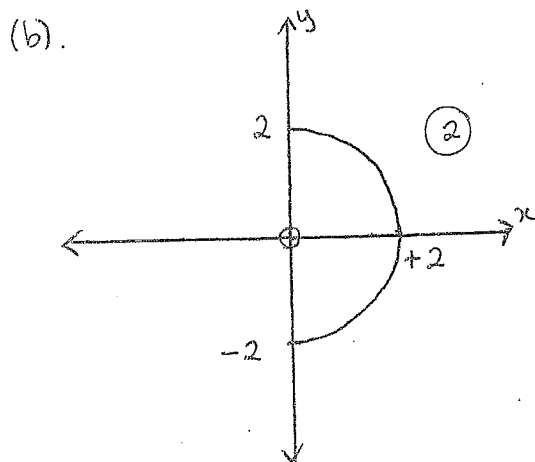
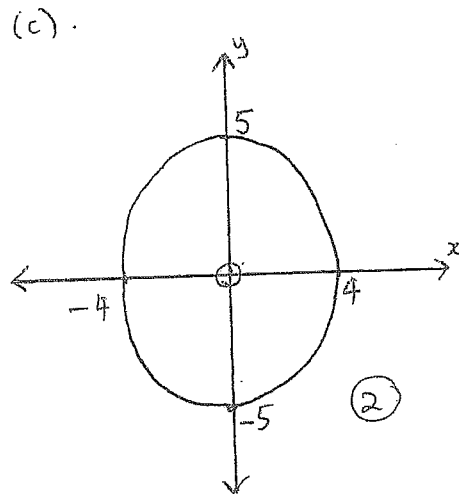
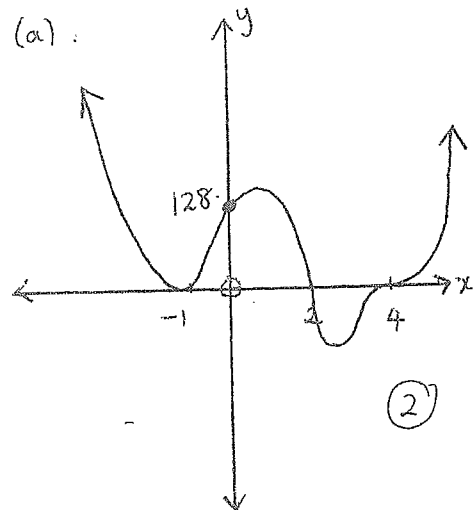
The particle P moves with constant speed, v m/s, in a horizontal circle about the midpoint of AB so that both pieces of string experience a tension.

The tension in AP is T_1 and the tension in BP is T_2 . The acceleration due to gravity is $g \text{ ms}^{-2}$.

- Copy the diagram and mark on it all the forces acting on P. 2
- Find the length of the radius of motion OP. 1
- Resolve the forces on P horizontally and vertically. 2
- Find the tension in each part of the string in terms of m , v and g . 4
- Show that $v > \frac{4\sqrt{3g}}{15}$. 3

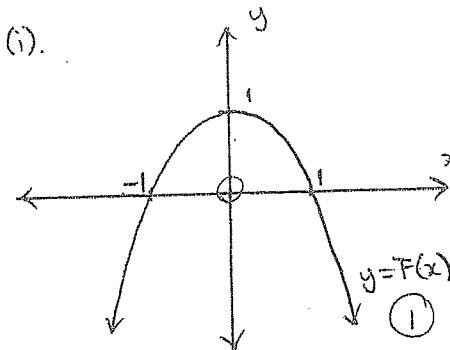
2010 (Nov '09) - Extension 2 - TASK 1: SOLUTIONS

Question 1

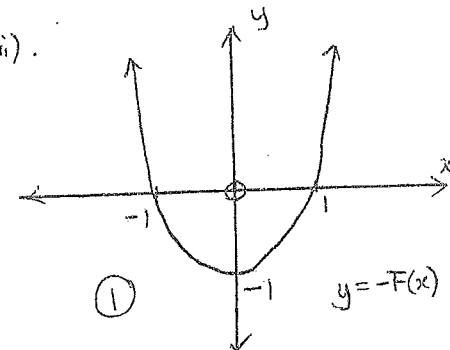


QUESTION 2

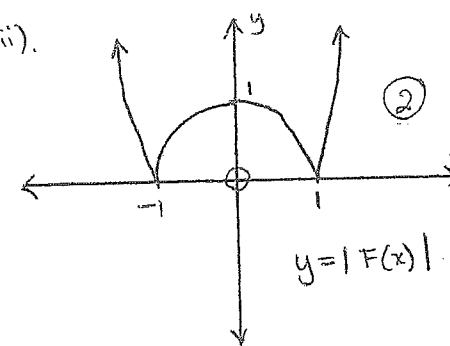
(a). (i).



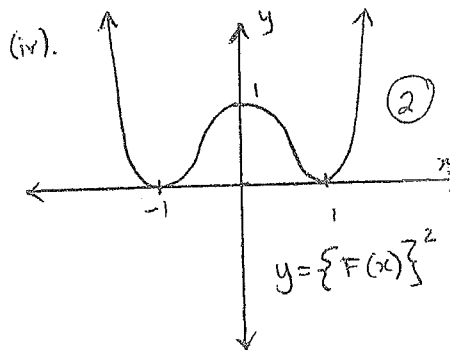
(ii).



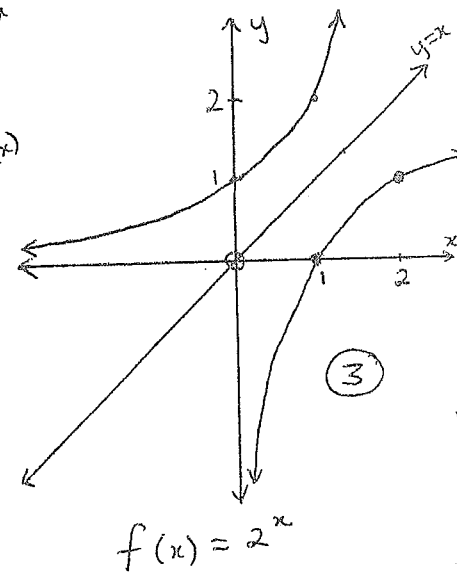
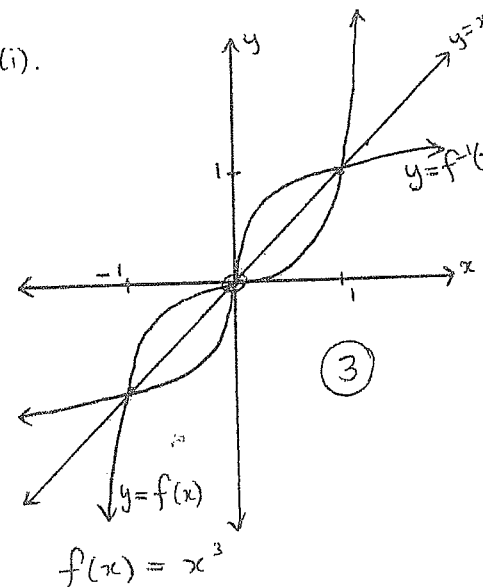
(iii).



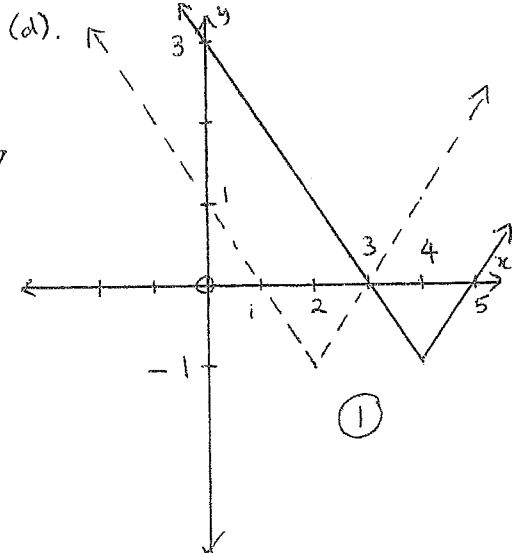
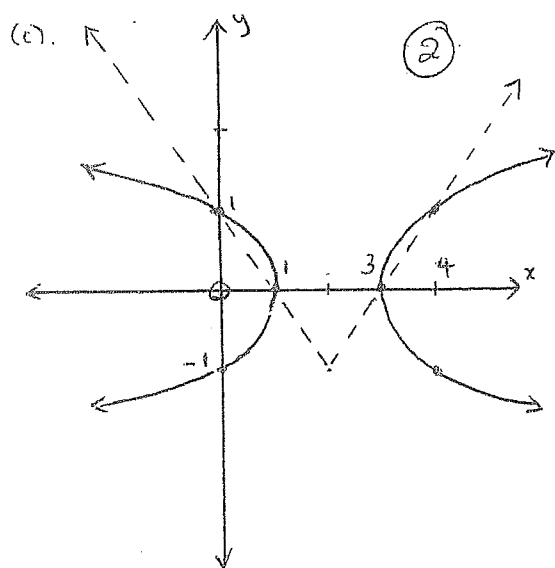
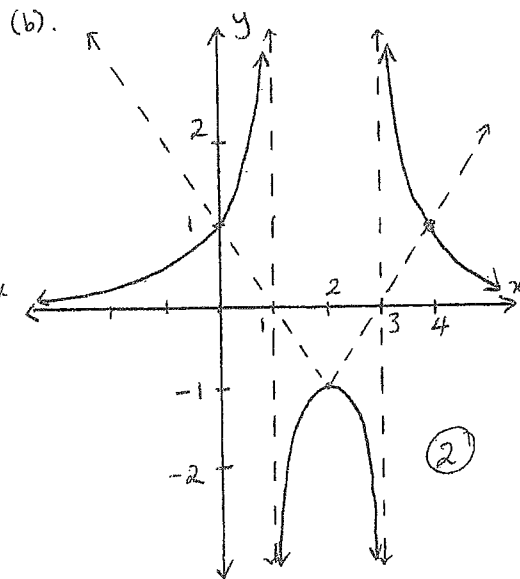
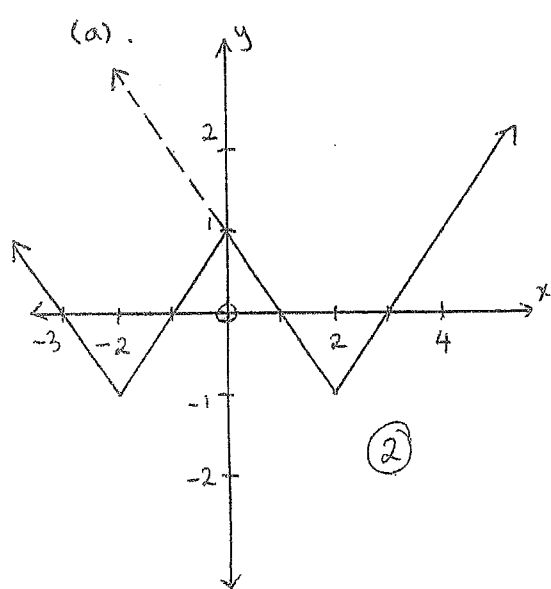
(iv).



(b). (i).

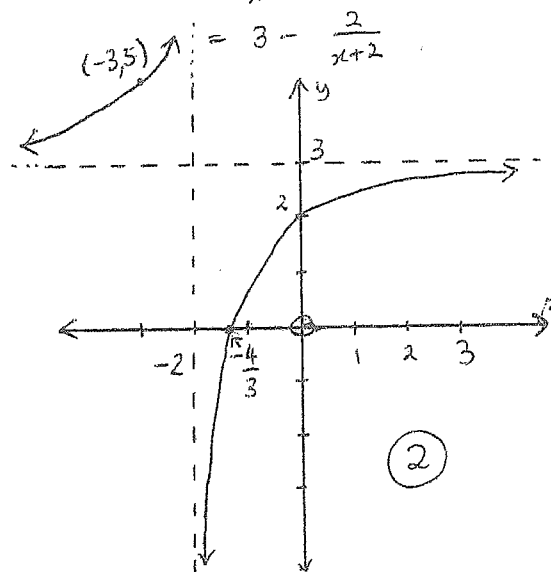


QUESTION 3

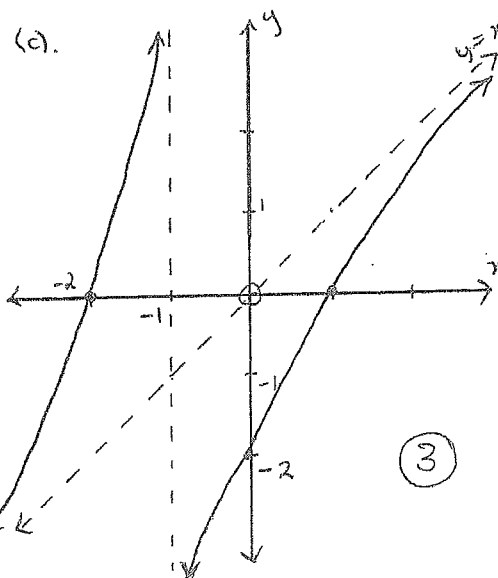
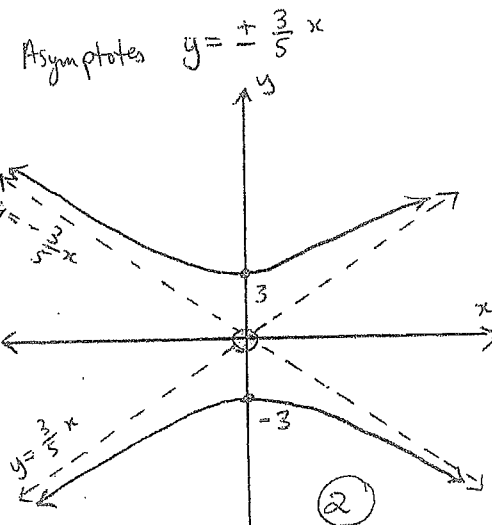


QUESTION 4

(a). $y = \frac{3x+6}{x+2} - \frac{2}{x+2}$

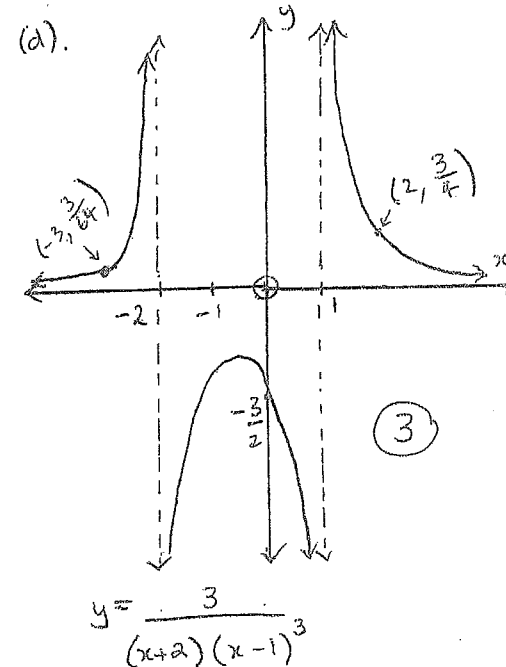


(b). $(5y+3x)(5y-3x) = 225$



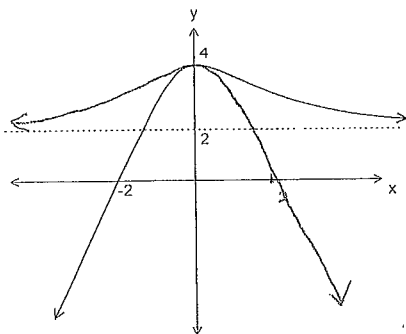
$y = x - \frac{2}{x+1}$

$$= \frac{x^2 + x - 2}{x+1} = \frac{(x+2)(x-1)}{x+1}$$

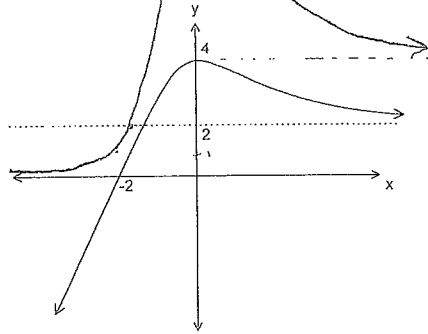


QUESTION 5

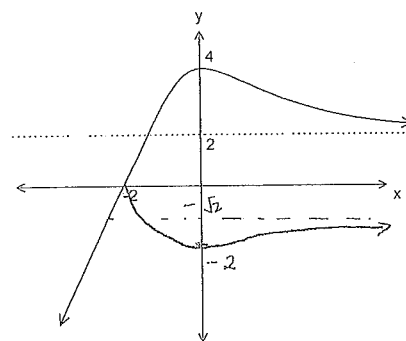
(a). $y = f(-x)$ ①



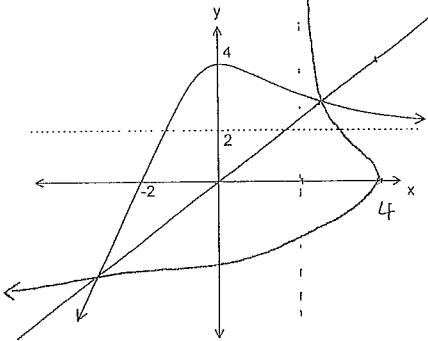
(b). $y = 2f(x)$ ②



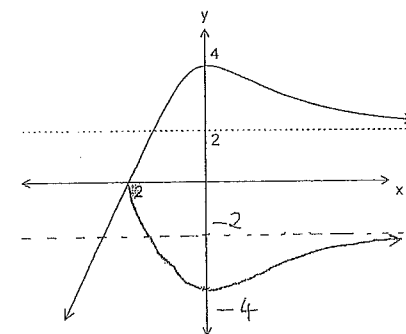
(c). $y = -\sqrt{f(x)}$ ②



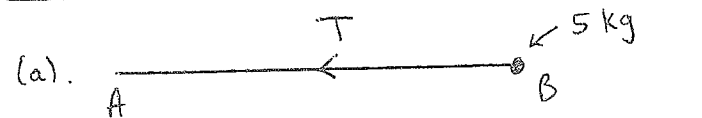
(d). $y = f^{-1}(x)$ ②



(e). $|y| = f(x)$ ①



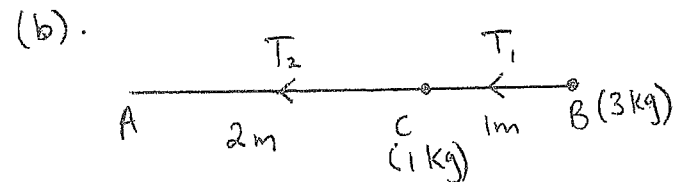
QUESTION 6



$$\begin{aligned} \omega &= 45 \text{ revs/min} \\ &= \frac{45 \times 2\pi}{60} \text{ rad/s} \\ &= \frac{3\pi}{2} \text{ rad/s} \end{aligned}$$

$$\begin{aligned} T &= mr\omega^2 \\ &= 5 \times 3 \times \left(\frac{3\pi}{2}\right)^2 \\ &= \frac{135\pi^2}{4} \text{ N} \end{aligned}$$

③



$$\begin{aligned} T_1 &= mr\omega^2 \\ &= 9\omega^2 \end{aligned} \quad \begin{aligned} T_2 &= 2 \times 1 \times \omega^2 + T_1 \\ &= 11\omega^2 \end{aligned}$$

$$11\omega^2 = \frac{135\pi^2}{4}$$

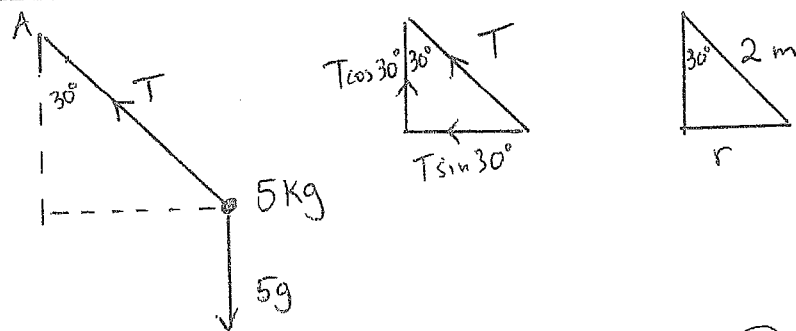
$$\therefore \omega = \sqrt{\frac{135\pi^2}{44}} \text{ rad/s}$$

$$= \sqrt{\frac{135\pi^2}{44}} \times \frac{60}{2\pi} \text{ rev/min.}$$

$$\omega \approx 52.5 \text{ rev/min.}$$

⑤

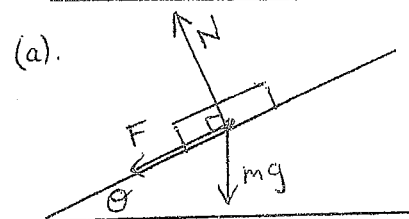
QUESTION 7



(a). Vert. $T \cos 30 = 5g$
 $T = \frac{5g}{\cos 30} = \frac{100}{\sqrt{3}}$
 $T \doteq 57.7 \text{ N}$

(b). $T \sin 30 = \frac{mv^2}{r}$ $\sin 30 = \frac{r}{2}$
 $\therefore r = 1$
 $v^2 = \frac{r T \sin 30}{m}$
 $= \frac{100}{\sqrt{3}} \times \frac{1}{2} \times \frac{1}{5}$
 $= \frac{10}{\sqrt{3}} \therefore v = \sqrt{\frac{10}{\sqrt{3}}} \doteq 2.40 \text{ m/s}$

QUESTION 8



(b).
Hor.
 $N \sin \theta + F \cos \theta = \frac{mv^2}{r}$
Vert.
 $N \cos \theta - F \sin \theta - mg = 0$
i.e. $N \cos \theta - F \sin \theta = mg$

(c). $N \sin \theta \cos \theta + F \cos^2 \theta = \frac{mv^2}{r} \cos \theta$ (1)
 $N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta$ (2)
(1) - (2) $F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$
 $N \sin^2 \theta + F \sin \theta \cos \theta = \frac{mv^2}{r} \sin \theta$ (3)
 $N \cos^2 \theta - F \sin \theta \cos \theta = mg \cos \theta$ (4)
(3) + (4) $N = \frac{mv^2}{r} \sin \theta + mg \cos \theta$

(d). $r = 150 \text{ m}$ $F = 0$ $v = 120 \text{ km/h} = 33\frac{1}{3} \text{ m/s}$ $\theta = ?$

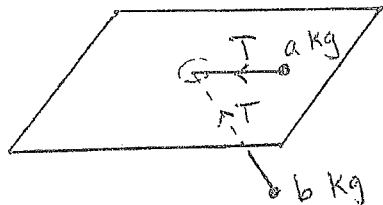
$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

$$0 = m \left(\frac{(33\frac{1}{3})^2}{150} \cos \theta - 10 \sin \theta \right)$$

$$\therefore \tan \theta = \frac{(33\frac{1}{3})^2}{150 \times 10} \therefore \theta \doteq 36^\circ 32'$$

No Half marks Total 10

QUESTION 9



$$a \text{ kg} \rightarrow w \text{ rad/s}$$

$$b \text{ kg} \rightarrow 2w \text{ rad/s}$$

$$a:b = 5:2$$

$$\therefore 2a = 5b$$

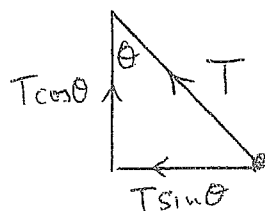
$$\text{or } b = \frac{2a}{5}$$

(a) kg

$$T = mrw^2$$

$$= arw^2$$

(b) kg



$$T \sin \theta = mrw^2$$

$$T \times \frac{R}{l} = \frac{2a}{5} \times R \times (2w)^2$$

$$\frac{arw^2}{l} = \frac{8aw^2}{5}$$

$$\therefore \frac{r}{l} = \frac{8}{5}$$

$\therefore \frac{8}{13}$ of string lies above the table

(7) marks

Notes

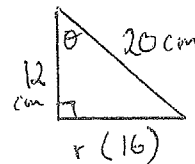
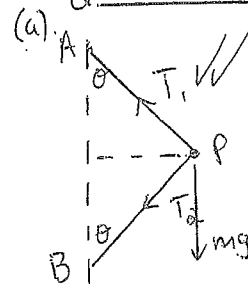
• No half marks

• Methods of soln varied

egs - put $a=5$, $b=2$

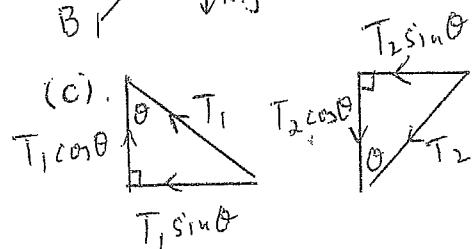
- Using $T_{\text{below}} = m l w^2$

QUESTION 10



$$r = \sqrt{20^2 - 12^2}$$

$$\therefore r = 16 \text{ cm or } 0.16 \text{ m}$$



$$\text{Hor.} \quad T_1 \sin \theta + T_2 \sin \theta = \frac{mv^2}{r}$$

$$\text{Vert.} \quad T_1 \cos \theta - T_2 \cos \theta - mg = 0$$

$$\text{i.e. } T_1 \cos \theta - T_2 \cos \theta = mg$$

$$(d) \quad \sin \theta = \frac{16}{20} \text{ or } \frac{4}{5} \quad \cos \theta = \frac{12}{20} \text{ or } \frac{3}{5}$$

$$(T_1 + T_2) \sin \theta = \frac{mv^2}{0.16} \quad (T_1 - T_2) \cos \theta = mg$$

$$T_1 + T_2 = \frac{125mv^2}{16} \quad (1) \quad T_1 - T_2 = \frac{5}{3} \quad (2)$$

$$(1) + (2) \quad 2T_1 = \frac{125mv^2}{16} + \frac{5mg}{3} \quad \therefore T_1 = \frac{125mv^2}{32} + \frac{5mg}{6}$$

$$(1) - (2) \quad 2T_2 = \frac{125mv^2}{16} - \frac{5mg}{3} \quad \therefore T_2 = \frac{125mv^2}{32} - \frac{5mg}{6}$$

$$(e) \quad T_2 > 0 \quad \checkmark \quad \frac{125mv^2}{32} - \frac{5mg}{6} > 0 \quad \checkmark \quad \text{No c.f.p. here}$$

$$\frac{125v^2}{32} > \frac{5g}{6} \quad \checkmark \quad v > \sqrt{\frac{9}{75}}$$

$$v^2 > \frac{16g}{75} \quad \checkmark \quad v > \frac{4\sqrt{g}}{5\sqrt{3}} \quad (3)$$

• No half marks. Total 12 marks i.e. $v > \frac{4\sqrt{3g}}{15}$